

XueYan Zhang

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SUMMARY

- 3+ years AI NPU compiler with multiple top-tier publications (ACL, EMNLP, COLM) toward AI full-stack expertise
- Career goal: Technical expert/architect role balancing cutting-edge research with production impact
- Core strengths: rapid execution from concept to production; technical leadership across the AI stack

EXPERIENCE

- **Heterogeneous Compiler Lab, Huawei Technology Canada** Toronto, Ontario
Senior Compiler Engineer Sept 2022 - Present
 - **NPU Compiler Optimization for LLMs:**
 - * Spearheaded compiler optimizations in LLVM for featured networks, like Flash Attention and RMS norm, achieving 25% improvement in operator performance and a 10% reduction in compiled binary size.
 - * Served as the technical focal point for a 3-person team, leading communication and aligning technical progress with international counterparts.
 - * Key: peephole optimizations, instruction prefetch, custom loop fusion, and zero-overhead hardware loops.
 - **Unified Test Case Reduction Framework:**
 - * Architected an award-winning test case reduction framework by unifying 5 open-source algorithms and pioneering a novel, LLM-based approach to auto-generate test properties.
 - * This service is now used by 200+ developers and has boosted debugging efficiency by 50%, automatically minimizing a 10k-line test case to its 100-line root cause in under 30 minutes.
 - **Ahead-of-Time Constant Propagation:** Implemented constant identification algorithm to dynamically extract the parameter struct within the kernel graph during runtime. Enabled online compilation to swap binaries, achieving up to 12% faster execution. Contributed to benchmarking infrastructure.
 - **General Matrix Multiplication in MLIR:** Engineered a vectorization feature in MLIR to pattern match, transform, and optimize user codes to fully utilize the matrix computing unit. Supported double buffering.
 - **Student Intern Mentorship:** Mentored multiple software engineering interns on compiler projects, leading to the successful integration of their features into the production codebase and a final results presentation to headquarters.
- **UWaterloo CS136, Instructional Apprentice** Waterloo, Ontario
Clarified complex concepts in algorithms and data structures for students in weekly office hours. Sept 2021 - May 2022
- **UWaterloo SWAG Lab, Undergraduate Research Assistant** Waterloo, Ontario
Engineered benchmarking automation with tracking, quantifying, and visualizing Sept 2020 - Dec 2020
- **Imperial Oil, Upstream Facility Engineer Intern** Calgary, Alberta
Served as interim production site contact; completed 100+ facility engineering packages May 2018 - Apr 2019

PROJECTS

- **iMarkAi.com, Agentic bookmark with semantic search & web interactions** Nov 2025
 - Design, implement, and successfully published a commercial-grade bookmark extension on Chrome Web Store.
 - Architected intelligent search platform using Pinecone vector database and Google Gemini AI, achieving sub-second semantic retrieval across unstructured web data with 95%+ precision.
 - Implemented end-to-end Stripe billing for subscriptions and usage-based payments.
- **ArchiValley.AI, Web-based AI architectural renderer from 3D models or sketches** Jan 2026
 - Identified and solved the critical bottleneck in architectural design cycles, slow and expensive, by developing an AI-powered platform that reduces visualization time from hours to seconds.
- **Tarot: Starry Guidance, End-to-end mobile app development** Sept 2025
 - Led the end-to-end, full-stack development on the WeChat mini-program platform; launched within 1 month.
 - Utilized web scraping for data; implemented HTML/CSS for UI, a cloud database, and an integrated LLM for personalized interpretations of drawn cards.
 - Conducted market research and competitor analysis; **reached 500+ users in first 3 months.**

SKILLS & SERVICE

- **Programming:** C/C++,  Python, Bash,  Java/Kotlin,  Unix/Linux, **git** Git
- **Toolbox:** Huggingface, Pytorch, LLVM, MLIR, Bazel,  Docker, ClaudeCode/Antigravity/Cursor

EDUCATION

- **University of Waterloo**, *Master of Math in Computer Science* Waterloo, Ontario
Thesis Stream in Software Engineering; Avg. 93/100; Graduate Excellence Award *Jan. 2021 – Dec. 2022*
- **McMaster University**, *Bachelor of Engineering in Mechatronics Co-op* Hamilton, Ontario
GPA: 11.6/12; Dean's Honor List; Senate Scholarship *Sept. 2015 – Apr. 2020*

PUBLICATIONS

- **UORA: Uniform Orthogonal Reinitialization Adaptation in PEFT** *May 2025*
 - Investigated Parameter-Efficient Fine-Tuning (PEFT) methods, with a focus on LoRA and its variants.
 - Reproduced over 8 state-of-the-art artifacts; benchmarked performance across diverse tasks in Natural Language Processing (NLU, NLG, Instruction-tuning) and Computer Vision (ViT).
 - Proposed a novel method of interpolation-based reinitialization mechanism to achieve SOTA parameter efficiency, reducing trainable parameters by up to 15x compared to LoRA while maintaining competitive performance.
 - *Accepted to Annual Meeting of the Association for Computational Linguistics (ACL 2025)*
- **Tiny Budgets, Big Gains: Parameter Placement Strategy in PEFT** *Aug 2025*
 - Introduced FoRA-UA, a super-efficient method that uses a sparse Fourier low-rank approximation with a universal shared adaptor, achieving state-of-the-art performance with only 1-5% of LoRA's parameters.
 - *Accepted to Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*
- **Ethical and Robust Large Language Models** *Sept 2024*
 - Investigated the ethical risks of 3 advanced LLM reasoning frameworks (CoT, Self-Reflection, Multi-Agent Debate). By evaluating 6 LLMs on fairness benchmarks across 4 languages.
 - Quantified a critical trade-off between reasoning and safety, finding that reasoning frameworks can degrade model alignment, with Multi-Agent Debate causing the largest drop in reasoning safety, a 34.4% decrease on average.
 - *Accepted to 4th Workshop on Adverse Impacts and Collateral Effects of Artificial Intelligence Technologies at The ACM WEB CONFERENCE 2026*
- **Large Language Model is not a (Multilingual) Compositional Relation Reasoner** *Jan 2024*
 - Pioneered a comprehensive, multilingual evaluation of LLM compositional reasoning by creating and open-sourcing the MCR Benchmark. This new benchmark systematically assesses model performance across six distinct relation categories in five languages (English, Chinese, French, Japanese, and Korean), identifying and quantifying critical reasoning deficiencies in state-of-the-art models.
 - *Accepted to Annual Conference on Language Modeling (COLM 2024)*
- **Word Semantics Consistency in Language Models** *May 2024*
 - Quantified the internal-external knowledge mismatch for word semantics across Encoder, Decoder, and Encoder-Decoder LMs.
 - Probing internal states versus querying model outputs on 3 benchmarks (word similarity, Named Entity Recognition (NER), analogy) revealed that probes access significantly more accurate semantic knowledge, highlighting a key representational discrepancy.
- **On the Caching Schemes to Speed Up Program Reduction** *Jan 2022*
 - Master's Research Topic. Program reduction minimizes the failing test case while preserving the bug-triggering properties.
 - Proposed Refreshable Compact Caching (RCC) by leveraging the characteristics of monotonically decreasing subset in program reduction, which is a both computational and memory-efficient caching scheme for program reduction.
 - Effectively reducing 64% redundant property-test queries, accelerating up to 43% in reduction time, and minimizing memory footprint by 99.97% from 100+ GB (text-based) to 6 MB (RCC).
 - *Accepted to ACM Transactions on Software Engineering and Methodology (TOSEM 2024)*

PEER REVIEW & COMMUNITY SERVICE & MORE

- Reviewer, International Conference on Learning Representations (ICLR 2025)
- Delivered Prompt Engineering, Reinforcement Learning, and Fine-tuning Crash Courses
- PC hardware enthusiast for ultra-compact ITX builds